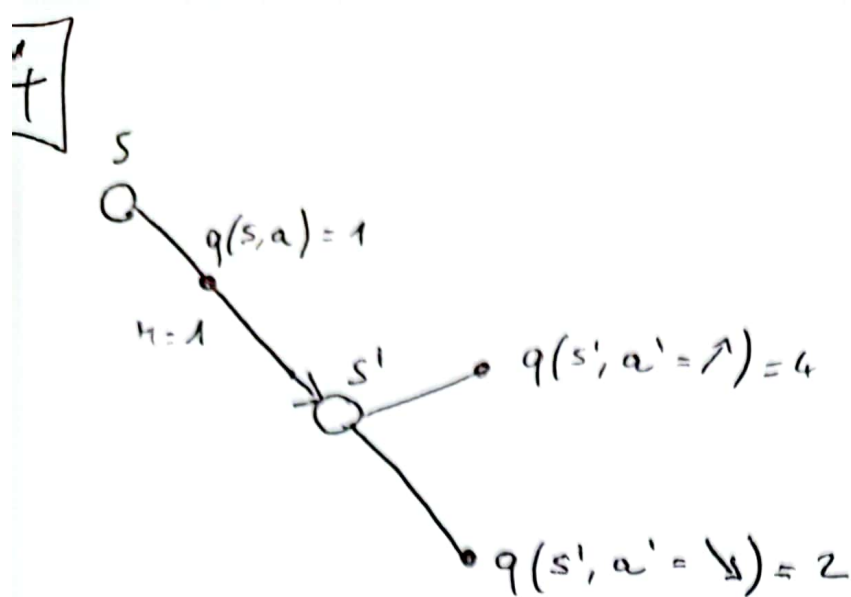




SARSA: $Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma Q(s', a') - Q(s, a)]$

$$Q(s, a) = 1 + 0.1 [1 + 0.5 \cdot 4 - 1] = 1.2 \quad \text{if } a' = 1$$

$$Q(s, a) = 1 + 0.1 [1 + 0.5 \cdot 2 - 1] = 1.1 \quad \text{if } a' = 2$$

Q-LEARNING: $Q(s, a) \leftarrow Q(s, a) + \alpha [R + \gamma \max_a Q(s', a) - Q(s, a)]$

$$Q(s, a) = 1 + 0.1 [1 + 0.5 \cdot 4 - 1] = 1.2$$

(INDEPENDENTLY FROM a' !)

EXPECTED SARSA: $\pi(s', a' = 1) = 2/3 \quad \pi(s', a' = 2) = 1/3$

$$Q(s, a) = 1 + 0.1 \left[1 + 0.5 \left(\frac{2}{3} \cdot 4 + \frac{1}{3} \cdot 2 \right) - 1 \right]$$

$$= 1 + \frac{1}{10} \cdot \frac{10}{3} = \frac{7}{3}$$

~~X~~